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A proposed global layout of carbon capture and storage in line with a 2 °C climate target

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A straightforward global layout of carbon capture, utilization and storage (CCUS) is imperative for limiting global warming well below 2 °C. Here, we propose a cost-effective strategy for matching carbon sources and sinks on a global scale. Results show 3,093 carbon clusters and 432 sinks in 85 countries and regions are selected to achieve 92 $GtCO_2$ mitigation by CCUS, 64% of which will be sequestered into sedimentary basins for aquifer storage and 36% will be used for CO_2-EOR (enhanced oil recovery). Of the identified source–sink matching, 80% are distributed within 300 km and are mainly located in China, the United States, the European Union, Russia and India. The total cost is ~0.12% of global cumulative gross domestic product. Of countries with CO_2-EOR, 75% will turn into profitable at the oil price over US$100 per barrel. These findings indicate our proposed layout is economically feasible. However, its implementation requires global collaboration on financial and technological transfer.

Carbon capture, utilization and storage (CCUS) has been regarded as a pivotal technological solution for limiting global temperature increases within 2 °C compared to the pre-industrial level[1,2]. It can provide a low-cost route to achieve substantial carbon reductions needed across key industries, such as power, steel, cement and chemicals[3,4]. The International Energy Agency (IEA) consistently highlights that CCUS would contribute to 94 Gt of cumulative CO_2 reduction in power generation, industrial processes and fuel switching by 2050 (nearly 12% of global mitigation) to achieve the 2 °C target[3]. However, the current development of CCUS has not reached the desired scale commensurate with the ambitions voiced a decade ago, with CO_2 captured being approximately less than 1% of the projection each year[5]. In particular, most current CCUS projects are still in the disordered planning stage and are located in only a few countries, such as the United States, Canada and China. It is obvious that CCUS technologies are suffering from the lack of an overall layout consistent with the 2 °C climate target[6].

However, most of the existing research regarding CCUS placed more emphasis on technological cost estimation[7–9], the theoretical modelling of CO_2 transport[10,11] at the project and regional levels, and precise point-to-point source and sink pairings[12,13] in the local area. Several tools are also coming online that can solve a CCUS problem for an entire county like the United States[11,14] or China[15], or a continent like Europe[16]. It can be seen that these studies are concentrating on a few technology-leading countries or regions, which cannot yet meet the 2 °C climate target. Consequently, it remains challenging to understand where the CO_2 emission reductions required by CCUS will come from, where they would go, and how the global carbon sources and sinks can be matched in a cost-effective solution at a global perspective.

The main obstacle for implementing the global deployment of CCUS is to identify the global distribution of large carbon sources that can capture CO_2 emissions and carbon sinks that can sequester CO_2. But there are no detailed data for such global source–sink matching in the current analysis and the large-scale point-to-point matching may lead to excessive costs. Considering that economies of scale can reduce the unit CO_2 avoidance cost[17], it would be more reasonable to investigate the large carbon cluster that is an area with multiple emissions sources inside and sharing the same pipelines. Furthermore, regarding the global assessment of the CO_2 storage potential, the announced potential in the existing mainstream research is limited to the national total amount without details on the spatial distribution across basins[18], which cannot support the design for the global CCUS source–sink matching. Although some countries have disclosed the spatial distribution and CO_2 storage potential of specific basins[7,19,20], due to differences in assessment methods, these results are not comparable. Thus, it is essential to apply a unified evaluation framework to assess the global storage potentials with high-spatial resolutions.

To address the above problems and limitations, we propose a high-level global collaborative layout of CCUS that can achieve the 92 $GtCO_2$ mitigation required by CCUS in line with the 2 °C target, filling in the gap of straightforward carbon source and sink matching solutions at the global scale and the minimum total investment required. Note that, out of the total 94 $GtCO_2$ mitigation by CCUS given by IEA[3], 2 Gt of cumulative CO_2 emissions from industrial processes is ignored due to data availability. Distinct from previous research that mainly focused on regional analysis, this study first constructs a gridded 1 × 1 km^2 global carbon source database and further identifies 4,220 carbon emission clusters that can be used to implement CCUS in priority. In addition, we assess the CO_2 storage potential of 794 basins that are suitable for CO_2 storage across the world, and disclose the map of global carbon sinks, contributing to solving the data gap that is preventing us from getting a better understanding of global CCUS potential. On the basis of the above

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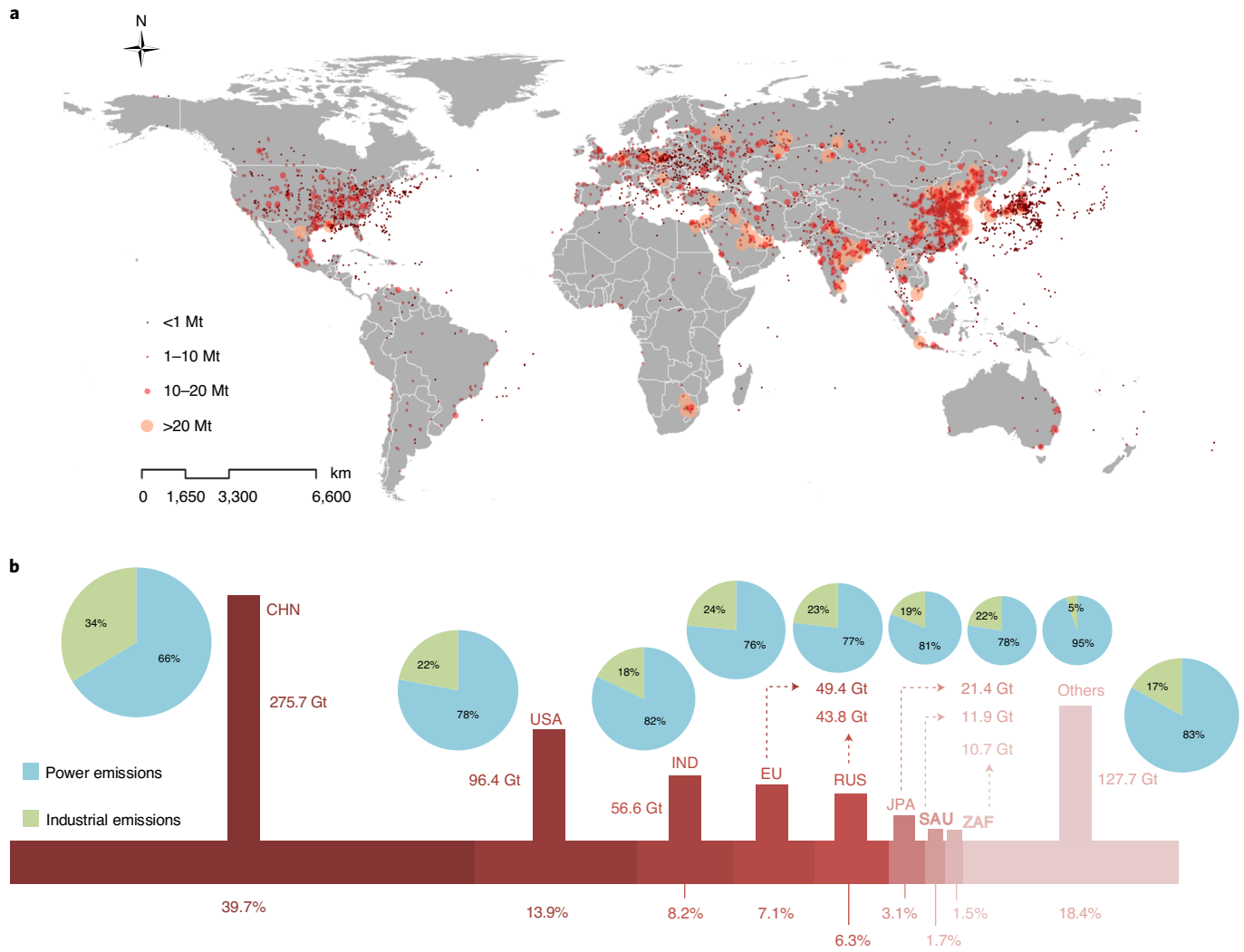


Fig. 1 | Spatial orientation of the worldwide carbon emission clusters considered in this study. a, The identified 4,220 carbon clusters. **b**, The cumulative carbon emissions from these clusters during 2020–2050. The percentage represents the cumulative emissions proportion of each region. The pie charts show the proportion of power emissions and industrial emissions (steel, cement and chemical) in each region. CHN, China; USA, the United States; IND, India; EU, European Union; RUS, Russia; JPN, Japan; SAU, Kingdom of Saudi Arabia; ZAF, The Republic of South Africa; Others, the rest of the world. Note that the global carbon source database is first established by integrating the $1\times1\,km^2$ global land-use pattern grids and multiple carbon emission grids, including 66,273 power plants[25], 20,491 non-power industrial plants of the United States, China and Europe[26,27], and global $10\times10\,km^2$ emissions grids of steel, chemical and cement industries for other countries[28]. The carbon emission clusters in this figure are obtained on the basis of global carbon source database. The historical gridded emissions of 2017 are used to predict the cumulative carbon emissions of those clusters by 2050 on the basis of the global CO_2 emissions amount projected by the IEA.

emissions and sinks inventory, this study goes beyond the previous research by developing a global CCUS source–sink matching model, named C^{3}IAM/GCOP (China's Climate Change Integrated Assessment Model/Global CCUS Optimal Planning), with the consideration of cost heterogeneity across carbon clusters in different countries. Finally, an operational and economically optimal CCUS deployment and the corresponding total investment from a global perspective that can meet the 2 °C target by 2050 are proposed.

Identification of global carbon emission clusters of CCUS

To better identify the grid positioning of carbon sources available for CCUS, we expand the existing emissions grids information (Emissions Database for Global Atmospheric Research, EDAGR) by constructing a more reliable gridded $1\times1\,km^2$ global carbon source database (Methods). On the basis of this, we introduce the notion of the carbon cluster for CCUS, which indicates an area consisting of adjacent grids with individual annual emissions exceeding 10,000 t and total annual CO_2 emissions exceeding 40,000 t (ref. [17]). In this way, multiple carbon sources within the carbon clusters can reduce the industrial chain risks and offer commercial synergies[21]. Finally, 4,220 suitable carbon emission clusters, mainly located in 87 countries, were identified (Fig. 1).

Our data show the cumulative CO_2 emissions of 4,220 carbon clusters by 2050 are 693.8 $GtCO_2$, of which three-quarters is emitted by power plants (518.8 $GtCO_2$). The number of carbon clusters with mixed emissions from power plants and other industries accounts for 47%; while the number of clusters with only power emissions and with only non-power emissions account for 24% and 29%, respectively. Of the clusters, 90% are with cumulative emissions higher than 0.1 $GtCO_2$. There are 47 countries in which almost all CO_2 suitable for carbon capture comes from power emissions. The distribution of 4,220 carbon clusters shows notable regional

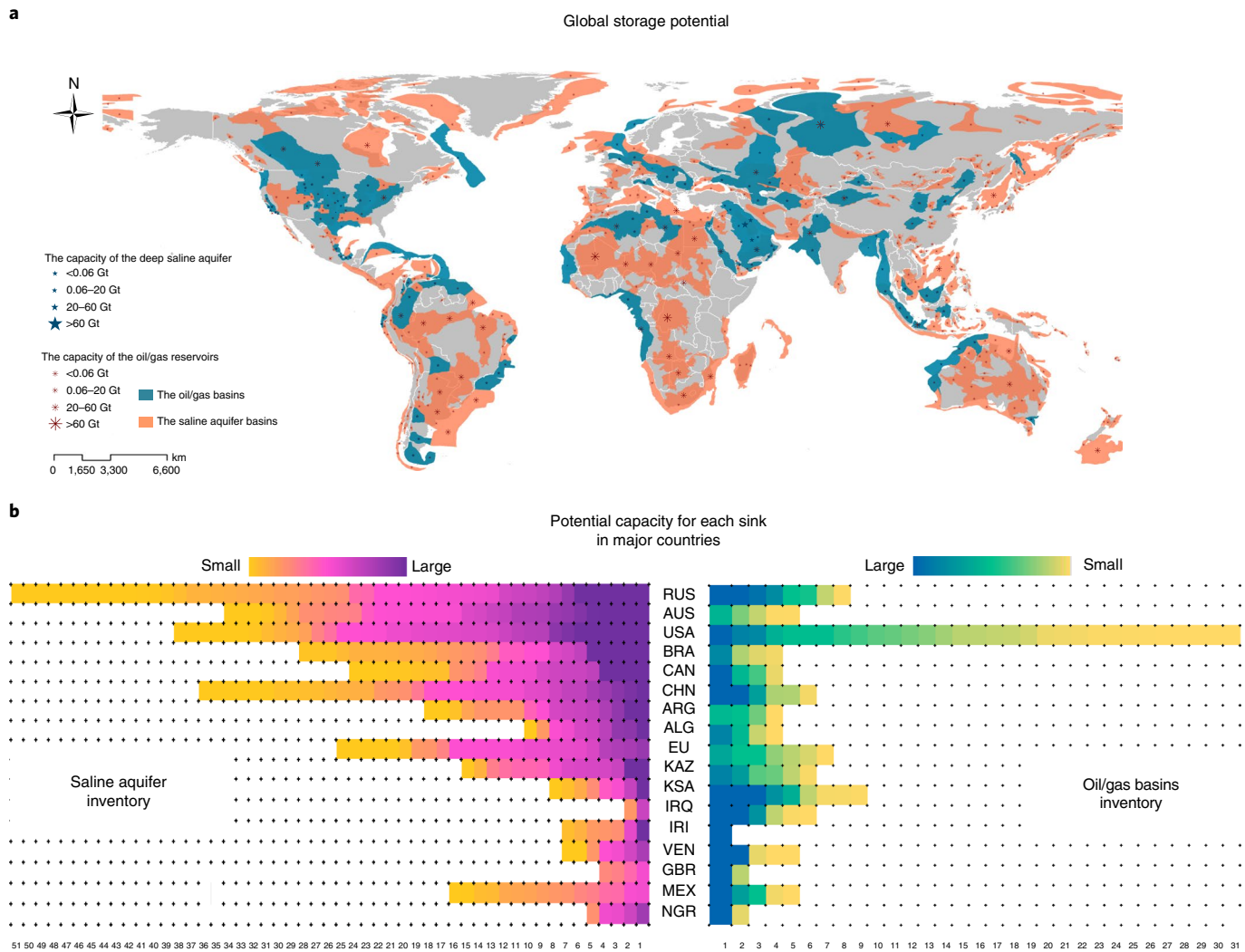


Fig. 2 | The global distribution of the effective CO_2 storage potential of different basins. a, The global distribution of oil reservoirs suitable for CO_2-EOR is highlighted in blue, and a larger pentagram indicates a greater magnitude of storage capacity. Similarly, the sedimentary basins for aquifer storage are displayed in pink with the superimposed star to reveal the amount of CO_2 storage. Basins with both offshore and onshore distributions are considered to only store CO_2 on onshore areas but the storage potential of their offshore distributions is still included. The storage potential of a transnational basin is allocated according to the proportion of the area occupied by the two countries. **b**, Illustration of the potential storage capacity for each sink in major countries. These countries are all in the top ten list regarding the total potential of each storage type. Each grid in **b** represents a carbon sink in the relative country and the colour of the grid indicates the storage potential of the carbon sink. An empty grid indicates that there is no carbon sink.

concentration. The number of carbon clusters and cumulative emissions of carbon clusters from 2020 to 2050 in China, the United States, India, the European Union, Russia, Japan, the Kingdom of Saudi Arabia and the Republic of South Africa account for approximately 81% and 84% of the global amounts, respectively. These countries are able to implement more CCUS in the future.

Assessment of global CO_2 geological storage potential

To ensure the comparability and consistency of this global-scale study, we adopt a unified method proposed by the US Department of Energy (US DOE)[22] to estimate the effective CO_2 geological storage potential of 794 onshore basins in 87 countries across the world, including 180 oil reservoirs for CO_2 enhanced oil recovery (EOR) and 614 sedimentary basins for aquifer storage (Methods).

Results show that the effective global geological storage potential of CO_2 is 2,082 Gt. The storage potential of the oil reservoirs with CO_2-EOR is approximately 168 $GtCO_2$ (8% in total), while sedimentary basins for aquifer storage have a significantly higher storage potential with 1,914 $GtCO_2$ (92%). Figure 2a illustrates the global distribution of the CO_2 storage potential. For oil reservoirs, the Kingdom of Saudi Arabia has the most abundant effective storage in oil reservoirs with CO_2-EOR, accounting for 17.9% of the total, followed by Russia (17%) and Iraq (12%). The top ten countries nearly contribute 78.2% of the global CO_2-EOR storage potential. In terms of sedimentary basins for aquifer storage, this study focuses on the global onshore sedimentary basins derived from the Geologic Provinces of the World[23]. Russia could contribute >12% of the global CO_2 storage capacity, while Australia could provide nearly 10%, followed by Brazil (7%). The top ten countries here could offer 65.3% of the global storage potential for aquifer storage. Countries in the top list regarding the total potential of each storage type are highlighted in Fig. 2b. Russia and the United States have the most carbon sinks available for aquifer storage and CO_2-EOR, respectively.

Optimum source–sink matching of CCUS for the 2 °C target

To explore the global cost-effective CCUS layout strategy that can achieve the 92 $GtCO_2$ necessary reduction required by CCUS, the

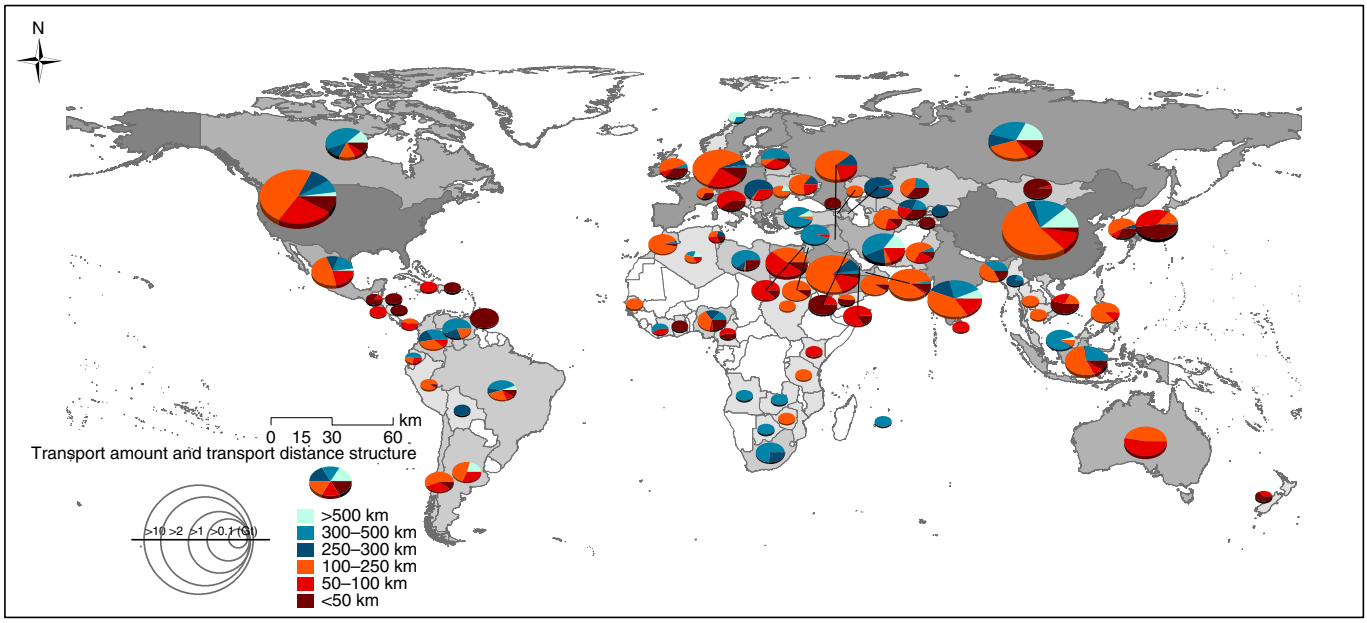


Fig. 3 | Transport amount and transport distance of the global cost-effective CCUS layout. The size of the pie charts represent the CO_2 transport volume of the 85 countries and regions participating in source–sink matching on a global scale. The composition of the pies indicate the proportion of the matches within different groups of transport distance.

C³IAM/GCOP is developed to match the above identified 4,220 carbon clusters and 794 usable basins distributed in 87 countries or regions (Methods).

Results show that 3,093 carbon clusters and 432 carbon sinks in 85 countries and regions are selected. A total amount of 58.61 $GtCO_2$ will be sequestered into sedimentary basins for aquifer storage, of which power carbon clusters and non-power industrial carbon clusters each supply about half. Another 33.39 $GtCO_2$ will be stored in the oil reservoirs for CO_2-EOR across the world, with 66% coming from CO_2 captured in the power sector. Of the source–sink matches, 80% are within an economically reasonable distance of 300 km defined by IPCC[18] and no pair of the matching exceeds 1,000 km (Fig. 3).

Global source–sink matching mainly locates in eight countries or regions (China, the United States, the European Union, Russia, India, the Kingdom of Saudi Arabia, Australia and Mexico), which undertake 76% of the required 92 $GtCO_2$ occurring between 2,399 carbon clusters and 197 carbon sinks (Fig. 4a–h). From the perspective of transport distance, >91% of the CO_2 can be matched to a suitable sink within 300 km in the United States, the European Union, Australia and the Kingdom of Saudi Arabia, which indicates that short-distance transport with small batches will be the main choice for the future CCUS layout of these four countries. In India and Mexico, 70–80% of the captured CO_2 would be transported within a reasonable distance <300 km, respectively, and the rest of their matches are almost distributed between 300 km and 500 km. For Russia and China, long-distance transport is relatively more than other countries. China has over 11% of the match amount exceeding 500 km and this proportion in Russia is even higher than 17%.

These eight countries or regions will undertake 1.86–26.68 $GtCO_2$ for CCUS (Fig. 4i). Specifically, there are a total of 1,430 carbon clusters suitable for CO_2 capture in China, 66% of which are selected in the proposed cost-effective CCUS layout strategy by 2050. These carbon clusters will require implementing CCUS to reduce 26.68 $GtCO_2$, 72% of which comes from non-power plants. Approximately one-fifth of the captured CO_2 needs to be used for EOR and the remaining will be sequestrated in deep saline aquifers. For the United States, 58% of the 841 carbon clusters contribute to reducing CO_2 emissions by 16.09 Gt, of which about 63% comes from the power sector. The EOR projects will undertake 5.27 $GtCO_2$ by 2050. In addition, CCUS will reduce 8.97 $GtCO_2$ from 507 carbon clusters in the European Union and 3.80 $GtCO_2$ from 163 carbon clusters in India, respectively. Most of the capture volume in these two countries is provided by the power sector and the CO_2 storage is dominated by sedimentary basins for aquifer storage (>75%). In contrast, Russia, the Kingdom of Saudi Arabia and Mexico mainly carry out CCUS through the CO_2-EOR projects. For Australia, almost all of the 1.86 $GtCO_2$ captured comes from the power sector, accounting for 98% of the total.

Overall costs for CCUS needed to meet the 2 °C target

Costs for CCUS deployment equal to the investment expenditures for CO_2 capture, transport and geological storage minus the benefits of the CO_2-EOR projects. Results show that our proposed global cost-effective CCUS layout strategy will demand nearly US$8.20 trillion of the financial investment in line with 2 °C targets and this will be offset by the benefits of CO_2-EOR projects, resulting in a total cost being US$5.76 trillion (in 2019 constant prices) up to 2050. The global average CO_2 price is ~US$62.65 t^{-1} CO_2.

More than 77% of the total cost occurs during the capture process, which is ~US$4.45 trillion up to 2050. The non-power sector will generate 39% of the capture cost and the power sector will account for 61%. Apart from the capture process, there are still US$3.25 trillion and US$0.50 trillion that will be needed for CO_2 transport and for CO_2 storage. Although the total cost of the current layout strategy is very high, it can still make US$2.44 trillion of benefits from the CO_2-EOR projects. If CO_2-EOR is not introduced into the strategy, the cost of global CCUS deployment would be much higher. Therefore, the development of the CO_2-EOR, rather than merely storing CO_2 underground, bring hope for the profitability and development of CCUS projects.

It is interesting to find that four countries (the Dominican Republic, Qatar, Kuwait and Haiti) can respectively earn US$2.49 million–US$4.66 billion from CO_2-EOR by fully offsetting

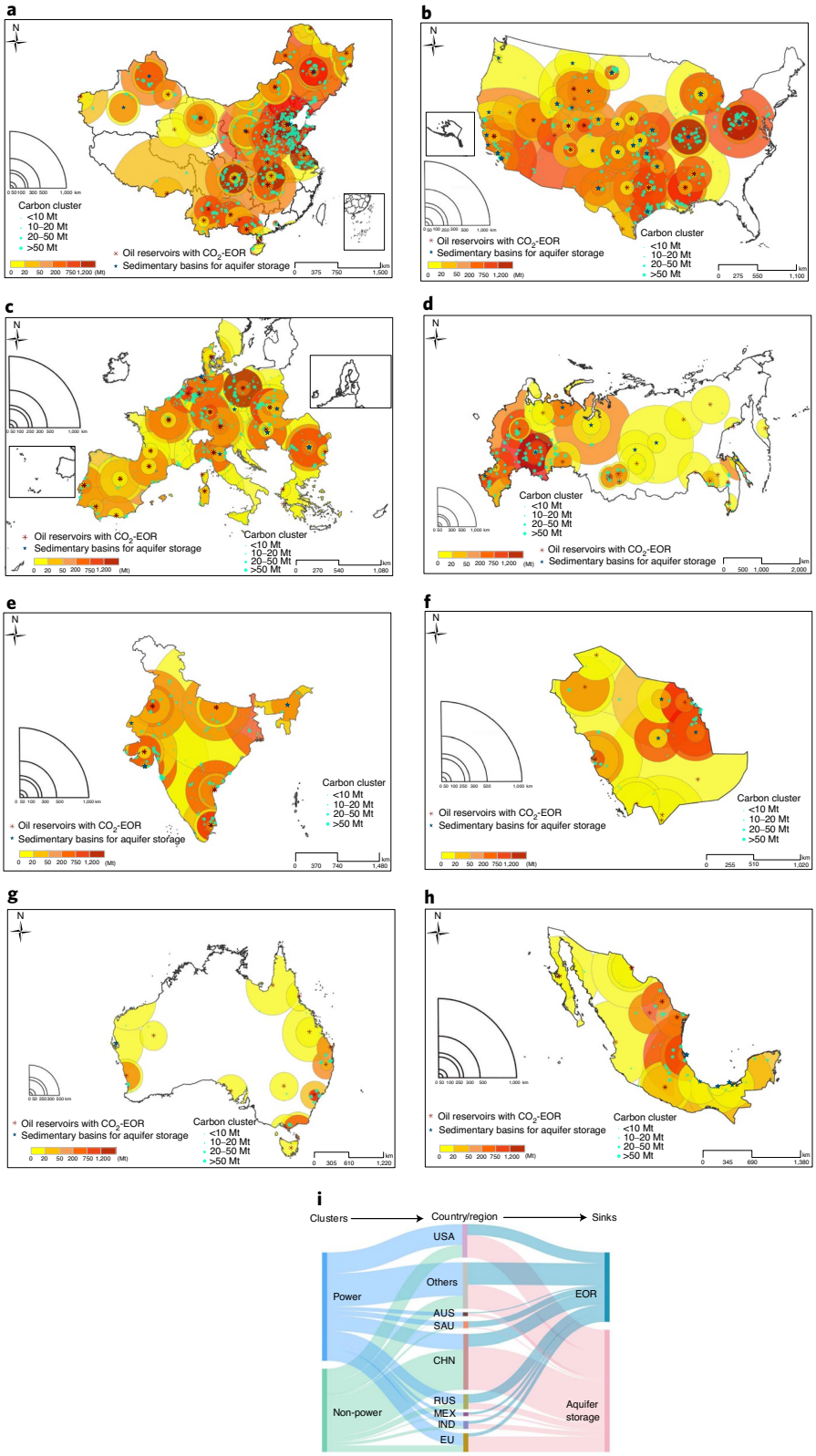


Fig. 4 | Optimized matching of the carbon emission clusters to the sinks until 2050 in eight countries and regions. a, China. **b**, The United States. **c**, The European Union. **d**, Russia. **e**, India. **f**, The Kingdom of Saudi Arabia. **g**, Australia. **h**, Mexico. **i**, CO_2 flow diagram. Here, we aggregate the individual source–sink match at the national or regional level and show the structure characteristics of the CCUS layout under different transportation distances. Each circle, centred on a sink, covers all clusters that match the sink. The colour indicates different transportation volumes of CO_2 captured, increasing from yellow to red. The radius of each circle represents different transport distances at <50 km, 50–100 km, 100–250 km, 250–300 km, 300–500 km, >500 km, respectively. Global source–sink matching mainly locates in China (26.7 $GtCO_2$), the United States (16.1 $GtCO_2$), the European Union (9.0 $GtCO_2$), Russia (7.2 $GtCO_2$), India (3.8 $GtCO_2$), the Kingdom of Saudi Arabia (3.3 $GtCO_2$), Australia (1.9 $GtCO_2$) and Mexico (1.8 $GtCO_2$). Related detailed results are shown in Supplementary Table 1.

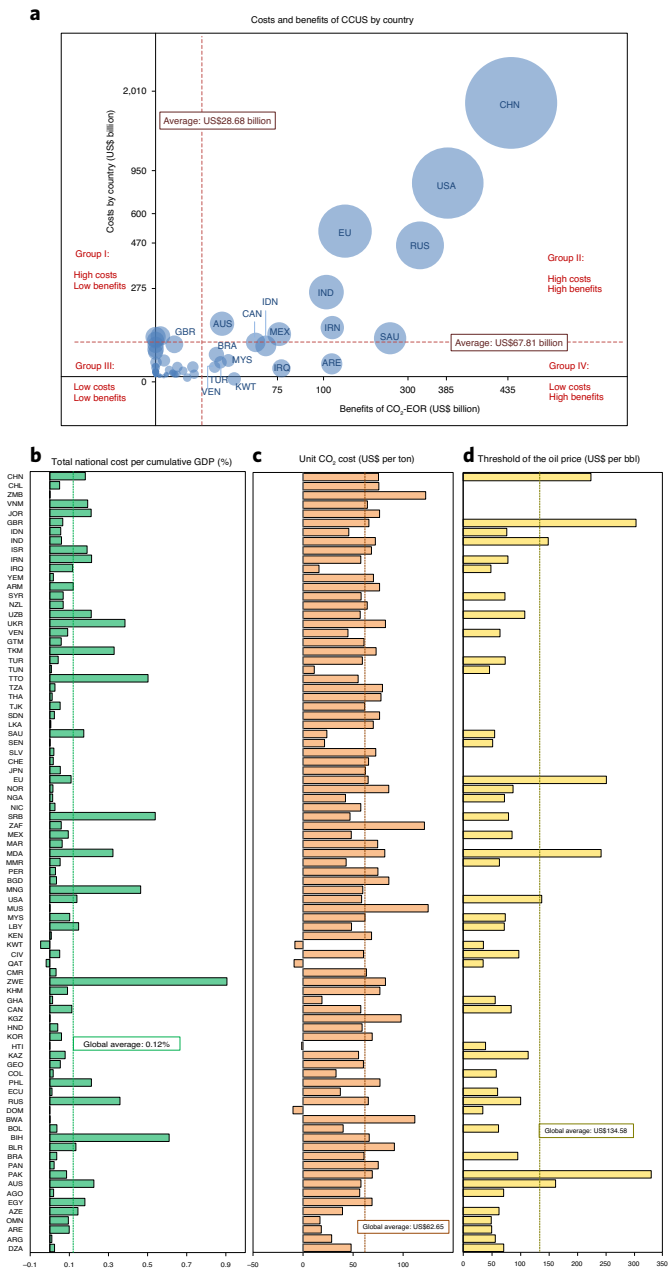


Fig. 5 | The economic performance of the cost-effective CCUS layout proposal across the world. a, Costs and benefits of implementing CCUS by country. The cost is calculated by subtracting the benefits of the CO_2-EOR projects from the investment expenditures for CO_2 capture, transport and geological storage. US$67.81 billion and US$28.68 billion are the average total cost and the average benefits of CO_2-EOR by country, respectively. The sizes of the bubbles indicate the amount of CO_2 stored in each country. **b**, The proportion of the total national cost to implement CCUS in their cumulative GDP until 2050 by country. The total global cost is ~0.12% of the global cumulative GDP until 2050. **c**, The unit CO_2 cost of CCUS by country. It is calculated by dividing the total national cost of implementing CCUS by the corresponding CO_2 emissions reductions by CCUS. **d**, The threshold of the oil price that makes CCUS deployment turnaround for each country/region in our global layout. The country abbreviations and cumulative GDP projections by 2050 are derived from the SSP database[24].

the cost of carbon capture, transport and storage; while the other 81 countries have to pay additional costs for CCUS deployment (Fig. 5). China, the United States and the European Union are near the top of the list, spending approximately US$2.01 trillion, US$0.94 trillion and US$0.58 trillion until 2050, respectively.

We categorize 85 countries and regions into four groups according to their total costs and benefits until 2050 (Fig. 5a). Japan, Ukraine and Egypt are in the group I that has higher costs due to low EOR benefits, ranging between US$74.50 billion and US$82.79 billion. These three countries undertake 3.7% of the global CO_2 storage. Countries such as China, the United States and the European Union belong to the group II that has higher EOR benefits but also higher costs. Countries in this group will contribute 79% of the global CO_2 reduction by CCUS and require the total cost of US$0.07 trillion–US$2.01 trillion. The group III has 64 countries with less CCUS deployments (only 10% of the global total), making their costs and benefits relatively small. There are also eight countries located in the group IV that could have the potential to make substantial EOR benefits in the future, within the range of total costs being –US$4.66 billion to US$62.49 billion. This shows that if oil prices rise or technology costs fall in the future, more and more countries in this group would profit by CO_2 use.

Regarding the unit cost obtained through dividing the total cost by total emission reduction, the national unit cost ranges between –US$10.02 and US$125.06 per ton of CO_2 (Fig. 5b). We further use the proportion of the total cost to implement CCUS in the projected cumulative gross domestic product (GDP) by 2050 under the SSP2 (Shared Socioeconomic Pathways[24]) scenario to evaluate the feasibility of different countries. Results show that four countries (the Dominican Republic, Qatar, Kuwait and Haiti) will increase their cumulative GDP by 0.0002% to 0.05% due to the development of CCUS until 2050. However, they are more vulnerable countries, indicating that upfront financial support and technological transfers from the developed countries are needed to help them promote CCUS before benefit. Conversely, for other major countries, developing CCUS requires a certain amount of the cumulative GDP. For Russia, China, the Kingdom of Saudi Arabia, the United States and the European Union, this expenditure would account for 0.36%, 0.18%, 0.17%, 0.14% and 0.11% of their cumulative GDP until 2050, respectively. In addition, in the context of our proposed strategy, the break-even threshold of the oil price for those countries deployed by CO_2-EOR varies from US$34.51 per barrel (bbl) to US$330.39 per bbl, with the global average threshold being US$134.58 per bbl. Twenty-five countries will profit when the oil price is ~US$75 per bbl theoretically and 75% of countries with CO_2-EOR will profit at the oil price over US$100 per bbl, indicating that CCUS is an economically promising technology for most of the countries. However, the actual break-even oil price to a large extent depends on the geological condition of each country.

Discussion

This paper contributes to the global cost-effective CCUS layout strategy for limiting the temperature rise below 2 °C. We develop an economically optimized approach named C^{3}IAM/GCOP on the basis of identifying 4,220 carbon clusters and evaluating 794 carbon sinks storage potentials around the world, providing a CCUS layout from a global perspective to achieve the CO_2 mitigation required by CCUS following IEA's projection.

Our cost-effective CCUS layout determines that a total amount of 58.6 $GtCO_2$ should be sequestered into 277 sedimentary basins for aquifer storage and 33.4 $GtCO_2$ should be sealed in 155 oil reservoirs for EOR across the world. Most matches occur in China (29%), the United States (17.49%), the European Union (9.75%), Russia (7.87%), India (4.13%), the Kingdom of Saudi Arabia (3.56%), Australia (2.02%) and Mexico (1.95%). The layout proposal also shows that 80% of the source–sink matching will be within a reasonable distance of 300 km, implying it is technologically feasible.

Our proposed CCUS layout strategy is able to address global warming in a cost-effective way. However, it requires the collaboration

of 85 countries and regions around the world. With their collaboration, implementing CCUS for achieving the 2 °C target only needs these 85 countries and regions to pay ~0.12% of the global cumulative GDP by 2050, which is considered to be affordable. In addition, four countries are able to obtain profit, such as Qatar and Kuwait, in which accelerating the deployment of CCUS should be encouraged. The other 81 countries need to pay a certain amount of costs, ranging from US$0.12 million to US$2.02 trillion. Capital and technology transfers from developed countries to less developed countries are needed for the global promotion of CCUS. In addition, the carbon clusters in the high-cost countries, such as Russia, Iran and the Kingdom of Saudi Arabia, are generally far from the sinks suitable for CO_2-EOR projects, resulting in their utilization rates of EOR potential being lower than that of the global average (21.54%). It is thus recommended that new carbon sources should be deployed near these carbon sinks in the future. More CO_2 utilization technologies should be introduced to offset the high cost of the entire process.

Due to the large differences in global geological structure, this study also conducts an uncertainty analysis on key geological parameters, including the oil formation volume factor, the density of CO_2 at storage conditions, the gross formation thickness and the total porosity in volume. The results show that the global CO_2 storage potential available for EOR may be within the range 124–300 $GtCO_2$; and the potential for deep saline aquifers may be within the range 888–5,126 $GtCO_2$. The effect of a 20% increase (high-cost scenario) or decrease (low-cost scenario) in the values of the initial unit costs of CO_2 capture, transport and storage on the global CCUS layout is further evaluated. The results show that under the high-cost scenario, deploying CCUS needs to pay US$7.38 trillion to achieve the 2 °C target with an average cost of US$80.26 per ton of CO_2. The distribution of CCUS efforts among countries around the world will not be changed evidently. The EOR project will undertake a reduction contribution of 30.84 $GtCO_2$. The total cost of global CCUS implementation under the low-cost scenario will decline to US$4.13 trillion, saving nearly US$17.77 per unit of CO_2 emission reduction. Lower transport and storage costs would make 33.36 Gt of captured CO_2 been transported to EOR projects. Considering the oil price is also an important factor affecting the EOR project benefits, we simulate the impact of changes in the oil price on the global layout of CCUS by increasing the oil price from US$55 to US$95 per bbl. The results indicate that high oil prices will help reduce the global cost of CCUS deployment and the unit cost will probably decrease to US$40.26 per ton of CO_2. Under the high oil price scenario, oil-rich countries, such as Iran, the Kingdom of Saudi Arabia and the United Arab Emirates will undertake more CCUS.

Taking into account each country's capability (storage capacity) and responsibility (carbon emissions), this study contributes to a global collaborative layout of CCUS in line with a 2 °C climate target. Because we intend to provide a high-level strategy, there are some uncertainties and limitations. For example, the difference of the geological conditions and detailed costs for the entire process of CCUS among countries is ignored due to the lack of the data. In addition, the uncertainty of oil prices and the explosion of breakthrough technologies will affect the results. Further research on a more sophisticated global layout of CCUS is needed.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41558-020-00960-0.

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Methods

Spatial identification of the global carbon sources available for capturing CO_2. To address the lack of data on global large-scale emission sources, this study identified carbon clusters suitable for CO_2 capture by establishing a $1\times1\,km^2$ carbon emission grids database that aggregated four industries, including the power generation, the iron and steel, the chemical and the cement industry. The data sources and process flows are described as follows and also presented in the Supplementary Fig. 1.

Specifically, for the power industry, we take the CO_2 emissions and location coordinates of 66,273 power plants in 2009 compiled by CARMA (Carbon Monitoring for Action) as the basic data of power industry point sources[25] and update the data by allocating the total CO_2 emissions of the power industry in 2017 calculated by the IEA according to the emissions proportion of each power plant given by CARMA. Regarding the non-power industries, we rasterize the coordinates and emissions of 20,491 non-power industrial point sources of the United States, China and Europe to obtain more reliable $1\times1\,km^2$ emissions grids[26,27]. We further use the EDGAR database to supplement and validate[28], which can provide $10\times10\,km^2$ CO_2 emissions grid-maps for non-power industrial activities (cement, chemical and iron and steel). We then use the global land-use grid map[29] to distribute the emissions of the non-power industry by excluding the grids in which forests, permanent wetlands, permanent snow and ice and water bodies are located. Subsequently, the emissions of a $10\times10\,km^2$ grid are downscaled to $1\times1\,km^2$ grids according to the remaining area after excluding inappropriate land-use types. Finally, the power plants emissions are rasterized into $1\times1\,km^2$ grids and further merged with the $1\times1\,km^2$ non-power emissions grids-map.

On the basis of the derived $1\times1\,km^2$ global carbon emissions grids database, we further define a $1\times1\,km^2$ area with CO_2 emissions greater than 10,000 t as a high-emission grid and concatenated the adjacent high-emission grids to a contiguous area and select the contiguous area with emissions exceeding 0.4 Mt as a carbon cluster. The carbon cluster in this study follows GCCSI's definition of a large CO_2 emission source[17]. We take the threshold as 0.4 Mt for all emitters, which are equivalent to the emissions of a medium-sized power plant.

Consequently, a total of 4,220 carbon clusters available in 87 countries globally are identified. In general, there would be no marked change in the physical location of large emission clusters within 30 yr, unless a force majeure event occurs, which is not within the scope of our discussion. Thus, the historical gridded emissions of 2017 are used to predict the cumulative carbon emissions of those clusters by 2050 on the basis of the global CO_2 emissions amount projected by the IEA, as shown in Fig. 1.

Assessment method of the global CO_2 storage potential. CO_2 can be geologically sequestered in oil and gas reservoirs, sedimentary basins for aquifer storage and unmineable coal areas[1]. This study only considers deep saline formations and CO_2-EOR in oil basins due to concerns about the safety of geological storage[30]. The sinks database in this study is based on the World Geologic Provinces database published by the United States Geological Survey[23]. The current assessment of CO_2-EOR potentials here only evaluated the CO_2 geological potentials of known oil fields (KWN_OIL) in the case of miscible flooding and near-miscible flooding. Therefore, this study does not consider undiscovered oil reservoirs and unconventional oil reservoirs such as ultra-high-pressure oil reservoirs or high sulfur oil reservoirs. Since the early 1990s, countries across the world have begun to assess the CO_2 geological storage potential at the national level[31,32] or for certain basins[33–35]. However, the existing assessment methods of CO_2 geological storage potential are quite different, leading to the great uncertainty of the assessment results[19,36,37]. To address this limitation, CSLF (Carbon Sequestration Leadership Forum)[37] and the US DOE[38], respectively, proposed two comprehensive evaluation systems of CO_2 geological storage. Goodman et al.[22] have reviewed that these two methods are essentially equivalent and are mainly applicable to evaluations at the national and basin scales. In comparison, the CSLF method requires more detailed geological parameters for evaluation, with residual gas saturation being directly considered. In addition, site-scale assessment methods are gradually being developed[1,39]. However, from a global perspective, it is difficult to obtain detailed geological parameters of all basins in various countries. Because the US-DOE methodology is intended to produce high-level estimates of potential CO_2 storage[22], this study selects the US-DOE framework to evaluate the CO_2 storage potentials of all 794 onshore basins around the world. Furthermore, to reduce the uncertainty of the evaluation, this study also estimates a spectrum of global storage potentials on the basis of three value levels of key geological parameters.

The global CO_2 storage potential in oil reservoirs with CO_2-EOR. This study follows the US-DOE methodology framework[22,40] to estimate the effective CO_2 storage potential in 180 oil reservoirs. Without considering the difference of CO_2 dissolution in water and crude oil, the effective CO_2 storage potential in oil reservoirs for CO_2-EOR on a global scale can be calculated as shown in equation (1):

$$G_{CO_2} = \mathrm{OOIP}\times\rho_{CO_2\mathrm{std}}\times E_{\mathrm{oil}} \tag{1}$$

where G_{CO2} refers to the CO_2 geological storage potential for CO_2-EOR; OOIP refers to the original oil-in-place; $\rho_{CO_2\mathrm{std}}$ refers to the density of CO_2 at standard conditions, usually acknowledged to be $1.977\,kg\,m^{-3}$ (ref. [41]); and E_{oil} indicates the CO_2 storage efficiency factor, which is generally set as 60% (ref. [42]).

The global CO_2 storage potential in sedimentary basins for aquifer storage. We assess the effective CO_2 storage potential of 614 sedimentary basins for aquifer storage, as follows:

$$V_{CO_2} = A\times\partial_A\times h\times\phi\times\rho_{CO_2}\times E \tag{2}$$

where V_{CO_2} is the effective CO_2 storage potential of sedimentary basins for aquifer storage; ρ_{CO_2} refers to the density of CO_2 at storage conditions, usually acknowledged to be $710\,kg\,m^{-3}$ (ref. [43]); A denotes the geographical area that defines the basin being assessed for CO_2 storage and ∂_A denotes the effective area ratio; h denotes the gross thickness of saline formations for which CO_2 storage is assessed within the basin defined by A; Φ denotes total porosity in volume defined by the net thickness; and E indicates the storage efficiency factor of CO_2, which is generally set as 0.05 (ref. [22]).

On the basis of the World Geologic Provinces, the actual area of all sedimentary basins worldwide was further calculated[23]. Following the research of Ecofys and Goodman et al., we characterize the global geological characteristics as three scenarios according to the value ranges of key parameters[40,42,43], including the oil formation volume factor, the density of CO_2 at storage conditions, the gross formation thickness, the total porosity in volume and so on. The detailed information on parameter values is shown in Supplementary Table 2. The main text only introduced the results under the medium level and the results under the other two scenarios can be found in Supplementary Table 3. In addition, the detailed CO_2 storage potential capacity and cumulative CO_2 emissions of identified carbon clusters in the United States, China and the European Union are listed in Supplementary Table 4.

C³IAM/GCOP model. A global source–sink matching model named C³IAM/GCOP, a submodule of C³IAM (refs. [44,45]), is developed to identify the optimum cost-effective CCUS layout strategy in line with 2 °C target among the above obtained 4,220 carbon clusters and 794 carbon sinks, with the goal of minimizing the total cost including the CO_2 capture, transport and storage (utilization) costs. Here, the carbon clusters would be directly connected to the carbon sinks, wherein a single cluster could simultaneously pair up with multiple sinks, and in turn, a single sink could be paired with multiple sources.

Objective function. A linear programming model is established for laying out global CCUS with transportation networks. The objective function is to minimize the total cost of global CCUS strategy during the planning period (2020–2050 in this study), as shown in equation (3), where f is the total cost of CO_2 capture, transportation and storage (utilization):

$$\begin{aligned}\min f = &\sum_{p=1}^{l}\sum_{i=1}^{n}\sum_{j=1}^{m}\left[\left(\mathrm{CC}_{pi}+\mathrm{TRC}_{pi,pj}\times D_{pi,pj}+\mathrm{SC}_{pj}\right)\times X_{pi,pj}\right]\\ &+\sum_{p=1}^{l}\sum_{i=1}^{n}\sum_{q=1}^{z}\left[\left(\mathrm{CC}_{pi}+\mathrm{TRC}_{pi,pq}\times D_{pi,pq}+\mathrm{SC}_{pq}-\mathrm{Rev}_{pq}\right)\times X_{pi,pq}\right]\end{aligned} \tag{3}$$

Here, p ($p=1,2,\ldots,l$) denotes country. Note that in the model, we practice the territory principle to match the global carbon clusters and carbon sinks, indicating that cross-border CO_2 transportation and matching are prohibited.

Carbon cluster is expressed by i ($i=1,2,\ldots,n$), where the power carbon clusters belong to the set of $\{i=1,2,\ldots,k\}$ and the non-power carbon clusters belong to the set of $\{i=k+1,k+2,\ldots,n\}$ ($k\le n$); carbon sinks of sedimentary basins for aquifer storage are expressed by j ($j=1,2,\ldots,m$) and the sinks for CO_2-EOR are expressed by q ($q=1,2,\ldots,z$).

CC_{pi} refers to the unit CO_2 capture cost of the ith carbon cluster in country p, which can be given by equation (4):

$$\mathrm{CC}_{pi} = \left(E_{pi-\mathrm{elec}}\times\mathrm{CC}_{pi-\mathrm{elec}}+E_{pi-\mathrm{nelec}}\times\mathrm{CC}_{pi-\mathrm{nelec}}\right)/E_{pi} \tag{4}$$

$$E_{pi} = E_{pi-\mathrm{elec}}+E_{pi-\mathrm{nelec}} \tag{5}$$

where $E_{pi-\mathrm{elec}}$ and $\mathrm{CC}_{pi-\mathrm{elec}}$ represent the CO_2 emissions amount and unit capture cost of the power generation emitters in the ith carbon cluster of country p; $E_{pi-\mathrm{nelec}}$ and $\mathrm{CC}_{pi-\mathrm{nelec}}$ represents the CO_2 emissions amount and the unit capture cost of non-power industrial emitters in the ith carbon cluster of country p. E_{pi} in equation (5) denotes the total CO_2 emissions of the ith carbon cluster in country p, which is the sum of the two types of the emitters. Furthermore, $\mathrm{CC}_{pi-\mathrm{elec}}$ and $\mathrm{CC}_{pi-\mathrm{nelec}}$ are formulated in the following description of the capture cost for the power/non-power emissions in detail, as shown in equations (27) and (33).

SC_{pj} or SC_{pq} refers to the unit CO_2 storage cost of the jth or qth sink in country p. $\mathrm{TRC}_{pi,pj}$ and $\mathrm{TRC}_{pi,pq}$ represents the unit CO_2 transportation cost from the ith carbon cluster to the jth or qth carbon sink in country p. Similarly, $D_{pi,pj}$ and $D_{pi,pq}$

represents the arc distance from the ith carbon cluster to the jth or qth carbon sink in country p. Supplementary Table 5 shows the detailed information on values and data sources of key parameters in the global CCUS source–sink matching model.

Rev_{pq} refers to the unit CO_2 benefit obtained from CO_2-EOR of the qth sink in country p, which is calculated by equation (6):

$$\mathrm{Rev}_{pq} = P_{\mathrm{oil}}/\mathrm{dpr}_{\mathrm{CO_2}} \tag{6}$$

where P_{oil} denotes the oil price, with the unit of US$ per bbl. $\mathrm{dpr}_{\mathrm{CO_2}}$ denotes the oil displacement rate, with a value of 4 (ref. [46]).

$X_{pi,pj}$ or $X_{pi,pq}$ is the CO_2 transport volume from the ith carbon cluster to the jth or the qth carbon sink in country p, which are also the decision variables in this model.

The constraints. Due to the capacity limitation, the amount of CO_2 captured by each cluster should be no more than the theoretical maximum value of the transport pipelines from the source, as shown in equations (7) and (8).

For any power emissions cluster i belonging to the set of $\{i=1,2,\ldots,k\}$,

$$\sum_{j=1}^{m} X_{pi,pj} + \sum_{q=1}^{z} X_{pi,pq} \le \eta_1 \times E_{pi} \tag{7}$$

For non-power industrial emissions clusters i belonging to the set of $\{i=k+1,k+2,\ldots,n\}$,

$$\sum_{j=1}^{m} X_{pi,pj} + \sum_{q=1}^{z} X_{pi,pq} \le \eta_2 \times E_{pi} \tag{8}$$

where η_1 and η_2 denote the maximum capture efficiency for the power plant point source and the non-power plant point source; E_{pi} denotes the total CO_2 emissions of the ith carbon cluster in country p.

The total capture amount of power plant point sources should be no less than the total CO_2 reduction amount of power sector that is required to achieve the 2 °C target, as shown in equation (9):

$$\sum_{p=1}^{l}\sum_{i=1}^{k}\sum_{j=1}^{m} X_{pi,pj} + \sum_{p=1}^{l}\sum_{i=1}^{k}\sum_{q=1}^{z} X_{pi,pq} \ge \mathrm{RRC_power}_A \tag{9}$$

where $\mathrm{RRC_power}_A$ denotes the necessary amount of CO_2 captured in the power sector through CCUS to achieve the 2 °C climate target, which is set as 52 $GtCO_2$ following IEA's results[3].

The total capture amount of non-power plant point sources should be no less than the total CO_2 reduction amount of non-power sectors that is required in line with a 2 °C target (equation (10)):

$$\sum_{p=1}^{l}\sum_{i=k+1}^{n}\sum_{j=1}^{m} X_{pi,pj} + \sum_{p=1}^{l}\sum_{i=k+1}^{n}\sum_{q=1}^{z} X_{pi,pq} \ge \mathrm{RRC_nonpower}_A \tag{10}$$

where $\mathrm{RRC_nonpower}_A$ denotes the necessary amount of CO_2 captured in the non-power sectors through CCUS to achieve the 2 °C climate target, which is set as 40 $GtCO_2$ following IEA's results[3].

In terms of aggregate quantity, for each country, the total amount of CO_2 captured from all domestic carbon clusters should not exceed the sum of the effective storage capacity of all sinks in the country, which is shown in equations (11) and (12). Furthermore, all the decision variables should be non-negative, as shown in equations (13) and (14):

$$\sum_{i=1}^{n} X_{pi,pj} \le Q_{pj} \tag{11}$$

$$\sum_{i=1}^{n} X_{pi,pq} \le Q_{pq} \tag{12}$$

$$X_{pi,pj} \ge 0 \tag{13}$$

$$X_{pi,pq} \ge 0 \tag{14}$$

where Q_{pj} denotes the maximum effective storage potentials of the jth carbon sink in country p; Q_{pq} denotes the maximum effective storage potentials of the qth carbon sink in country p.

Method for estimating the parameter of capture cost for the power emissions. To estimate the future dynamic capture cost of the power sector, that is, $\mathrm{CC}_{pi-\mathrm{elec}}$ in equation (4), we propose an optimal model with endogenous technological progress. Four CO_2 capture technologies suitable for the power sector are included: the precombustion capture of the integrated gasification combined cycle (IGCCC), postcombustion capture of supercritical pulverized coal (SPCC), precombustion capture of the natural gas combined cycle (NGCCC) and oxy-fuel combustion capture (Oxy).

The objective function of the model is to minimize the total capture cost of the power sector during the decision period (2020–2050), which is shown in equation (15):

$$\min\ \mathrm{TCC}_{\mathrm{elec}} = \sum_{t=2020}^{2050}\sum_{s=1}^{4}\Big[\mathrm{CC}_s(t)\times \mathrm{Cum}_s(t)\times H\times \mathrm{CF}_s \times \left(\mathrm{ER}_{s,\mathrm{ref}} - \mathrm{ER}_{s,\mathrm{CCS}}\right)\times (1+r)^{-(t-2020)}\Big] \tag{15}$$

where

$$\mathrm{CC}_s(t) = a_s \times \mathrm{Cum}_s(t)^{b_s} \tag{16}$$

where $\mathrm{CC}_s(t)$ represents the capture cost of the sth technology in year t (US$ per tCO_2); $\mathrm{Cum}_s(t)$ denotes the cumulative installed capacity of the sth technology in year t; and CF_s refers to the capacity factor of the sth technology. $\mathrm{ER}_{\mathrm{CCS}}$ and $\mathrm{ER}_{\mathrm{ref}}$ represent the carbon emission rate of the power plant with carbon capture and without carbon capture, respectively.

Equation (16) describes the learning curve of unit CO_2 capture cost for capture technology s, estimating the potential of technological cost reduction brought by the cumulative installed capacity. a_i and b_i are the parameters that can be obtained by equations (20)–(26).

To limit the temperature rise to well below 2 °C, the power sector is encouraged to undertake a 52 $GtCO_2$ reduction through carbon capture technologies[3]:

$$\sum_{t=2020}^{2050}\sum_{s=1}^{4} \mathrm{Cum}_s(t)\times H\times \mathrm{CF}_s\times \left(\mathrm{ER}_{s,\mathrm{ref}} - \mathrm{ER}_{s,\mathrm{CCS}}\right) \ge \mathrm{RRC_power}_A \tag{17}$$

Due to technical and engineering project constraints, the annual additional capacity of power plants with carbon capture is subject to a certain physical upper limit. Taking into account the global demand for clean power generation, the cumulative installed capacity of carbon capture and storage (CCS) power plants cannot exceed the maximum physical capacity of thermal power plants by 2050:

$$\sum_{s=1}^{4} \mathrm{Cum}_s(2050) \le S_{\max} \tag{18}$$

$$0 \le \mathrm{Cum}_s(t+1) - \mathrm{Cum}_s(t) \le A_s^{\max} \tag{19}$$

where $S_{\max}$ represents the maximum cumulative installed capacity of fossil fuel generation technologies in 2050[47] and $A_s^{\max}$ refers to the maximum annual recruited capacity of sth technology in year t (ref. [3]).

This study assumes that SPCC will show learning effects from 2020, while other technologies will have visible learning effects after 2030. Technology tends to mature once its market scale reaches 100 GW and the cost will no longer drop due to the increase in cumulative installed capacity[48].

The learning rate is the proportion of cost reduction when the output doubles. Here, we used the component-based learning curve theory and followed Li et al.[49] to describe the relationship between the capital/operation costs of a power plant and that of its various components:

$$\mathrm{LR}_{\mathrm{Ins,total}} = \lambda_1 \mathrm{LR}_{1,\mathrm{Ins}} + \lambda_2 \mathrm{LR}_{2,\mathrm{Ins}} + \ldots + \lambda_w \mathrm{LR}_{w,\mathrm{Ins}} \tag{20}$$

$$\mathrm{LR}_{\mathrm{OM,total}} = \alpha_1 \mathrm{LR}_{1,\mathrm{OM}} + \alpha_2 \mathrm{LR}_{2,\mathrm{OM}} + \ldots + \alpha_w \mathrm{LR}_{w,\mathrm{OM}} \tag{21}$$

where $\mathrm{LR}_{\mathrm{Ins,total}}$ denotes the learning rate of the total capital cost of a power plant. λ_w denotes the weight of the capital cost of the wth component in the total of the first CCS plant and $\mathrm{LR}_{w,\mathrm{Ins}}$ represents the learning rate of the capital cost of the wth component. Similarly, $\mathrm{LR}_{\mathrm{OM,total}}$ denotes the learning rate of the total operational costs of the first plant, while α_w and $\mathrm{LR}_{w,\mathrm{OM}}$ indicate the share and the learning rate of the wth component's operational cost, respectively.

The levelized cost of unit electricity generation (LCOE) of a power plant with carbon capture can be calculated by equation (22); and the unit CO_2 capture cost (CC) can be derived from LCOE, as shown in equation (23)[50]:

$$\mathrm{LCOE} = \frac{\mathrm{CRF}\times \sum_w^I + \sum_{w,\mathrm{OM}}^C + C_{\mathrm{fuel}}}{\mathrm{CF}\times H\times P} \tag{22}$$

$$\mathrm{CC} = \frac{(\mathrm{LCOE})_{\mathrm{CCS}} - (\mathrm{LCOE})_{\mathrm{ref}}}{\mathrm{ER}_{\mathrm{ref}} - \mathrm{ER}_{\mathrm{CCS}}} \tag{23}$$

where I_w and $C_{w,\mathrm{OM}}$ refer to the capital and operational costs of the wth component, respectively. C_{fuel} represents the annual fuel cost. P represents the rated power of the plant. CF represents the capacity factor. CRF represents the capital recovery factor. $\mathrm{ER}_{\mathrm{CCS}}$ and $\mathrm{ER}_{\mathrm{ref}}$ represent the carbon emission rate of the power plant with carbon capture and without carbon capture, respectively.

By combining the above formulas, the learning rate of LCOE and CC are formulated following their definitions by equations (24)–(26):

$$\begin{aligned} \mathrm{LR_{LCOE}} &= 1-\frac{\mathrm{LCOE}_d}{\mathrm{LCOE}_0} \\ &= 1-\frac{\mathrm{CRF}\times\left(1-\mathrm{LR_{Ins,total}}\right)\times I_0+\left(1-\mathrm{LR_{OM,total}}\right)\times C_{\mathrm{OM},0}+C_{\mathrm{fuel},d}}{\mathrm{CRF}\times I_0+C_{\mathrm{OM},0}+C_{\mathrm{fuel},0}}\times\frac{\mathrm{CF}_0}{\mathrm{CF}_d} \end{aligned} \tag{24}$$

$$\begin{aligned} \mathrm{CC}_d &= \frac{(\mathrm{LCOE})_{\mathrm{CCS},d}-(\mathrm{LCOE})_{\mathrm{ref},d}}{\mathrm{ER_{ref}}-\mathrm{ER_{CCS}}} \\ &= \frac{\left(1-\mathrm{LR_{COE,capture}}\right)\times\mathrm{COE_{CCS,0}}-\left(1-\mathrm{LR_{COE,ref}}\right)\times\mathrm{COE_{ref,0}}}{\mathrm{ER_{ref}}-\mathrm{ER_{CCS}}} \end{aligned} \tag{25}$$

$$\mathrm{LR_{CC}} = 1-\frac{\mathrm{CC}_d}{\mathrm{CC}_0} \tag{26}$$

where the subscript 0 indicates the initial value of the variable and d indicates the value when the cumulative installed capacity is doubled. The results of the different learning rates of four capture technologies can be seen in Supplementary Table 6, with learning curves in Supplementary Fig. 2.

Finally, according to our results of the total CO_2 capture cost of the power sector between 2020 and 2050, we estimated the average unit cost of the power carbon cluster in the source–sink matching model as follows:

$$\mathrm{CC_{elec}} = \mathrm{TCC_{elec}}/\mathrm{RRC_power}_A \tag{27}$$

Method for estimating the parameter of capture cost for the non-power emissions. To estimate the future dynamic capture cost of the non-power sectors, that is, $\mathrm{CC}_{pi-\mathrm{nelec}}$ in equation (4), we proposed a capture cost prediction model for the non-power carbon clusters. Considering the different configuration of equipment used in different plants and gas mixtures from various emitters, cost studies of CCUS in the industrial sector, unlike the power sector, are very limited and are very difficult. Therefore, due to the lack of available data, this study focused only on industries of iron and steel, cement, chemical and bio-energy with carbon capture and storage (BECCS).

Here, the global deployment rate of carbon capture equipment in non-power plants in year t is following an S-shaped curve (equation (28)):

$$\mathrm{tp}(t) = \exp\left(-\left(\left(U-U\left(\frac{t-\mathrm{SD}}{2050-\mathrm{SD}}\right)\right)^2\right)\right)\times\mathrm{tp_{max}} \tag{28}$$

where U represents the constant capture rate and SD represents the year when CCS was first installed. The variable $\mathrm{tp_{max}}$ denotes the maximum deployment rate of plants with carbon capture in the industry.

The cost learning curve of industrial carbon capture technology is estimated by $\mathrm{lc}(t)$ in equation (29), indicating that with an increase in the number of factories with CCUS, the CO_2 capture cost is reduced through technical learning. Moreover, the learning effect is suggested to decrease every 5 years:

$$\mathrm{lc}(t) = (1-\mathrm{CR})\times\mathrm{lc}(t-5)\forall\mathrm{tp}(t)>\mathrm{CR_{min}} \tag{29}$$

where $\mathrm{CR_{min}}$ represents the threshold of cost reduction; that is, the cost starts to decline when the industrial CCS deployment rate falls below this threshold. CR, is a constant value being 25% (ref. [51]), indicating there would be a 25% reduction in technology deployment rate every 5 years to reduce the learning effect after reaching the threshold $\mathrm{CR_{min}}$.

The increase in the cost of raw materials and industrial equipment directly affects the cost of the equipment and materials related to CCUS technologies. The annual additional cost of se−th sector ($\mathrm{ec_{se}}(t)$) is described by the initial cost of industrial technology ($\mathrm{CI_{se}}$) and the Chemical Engineering Plant Cost Index (CEPCI(t)). Here, se stands for the four sectors mentioned above: iron and steel, cement, chemical and BECCS. The initial unit capture costs of different non-power industries in this study are further described in Supplementary Note 1. The specific value of CEPCI(t) is shown in the Supplementary Fig. 3.

$$\mathrm{ec_{se}}(t) = \mathrm{CI_{se}}\times\mathrm{lc_{se}}(t)\times\mathrm{CEPCI}(t) \tag{30}$$

The total CO_2 capture cost of CCUS in non-power sectors is calculated as follows:

$$C_{\mathrm{nele}}(t) = \sum_{\mathrm{se}}(\mathrm{CAP_{se}}\times\mathrm{ec_{se}}(t)) \tag{31}$$

$$\mathrm{TCC_{nele}} = \sum_{t=\mathrm{SD}}^{2050} C_{\mathrm{nele}}(t)\times(1+r)^{-(t-2020)} \tag{32}$$

where $C_{\mathrm{nele}}(t)$ denotes the CO_2 capture cost of CCUS in the non-power sectors in year t and $\mathrm{CAP_{se}}$ denotes the corresponding amount of CO_2 captured. $\mathrm{TCC_{nele}}$ refers to the total CO_2 capture cost of CCUS in all non-power sectors by 2050.

Similar to the power sector, according to the calculated total capture cost of the non-power sectors, we estimate the average unit cost of the power carbon cluster in the source–sink matching model as follows:

$$\mathrm{CC_{nelec}} = \mathrm{TCC_{nelec}}/\mathrm{RRC_nonpower}_A \tag{33}$$

Data availability

The data that support the plots within this paper and other findings of this study are available from the corresponding author upon reasonable request.

Code availability

The codes that support the methods of this study are available from the corresponding author upon reasonable request.

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Author contributions

Y.-M.W., L.-C.L. and B.Y. conceived the study and performed the analysis. J.-N.K. and P.W. analysed the data and implemented the model. Q.L. and J.-J.H. contributed to the data collection and processing. S.-F.H. contributed to preparing the figures. Q.-M.L. and H.L. worked on the review and editing. All authors approved and contributed to writing the paper.

Competing interests

The authors declare no competing interests.

Additional information

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